

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 2.3 — Planning and Configuring Network Resources

GCP Associate Cloud Engineer

Study Notes Study Notes April 2026

Exam Relevance

This topic is part of **Section 2: Planning and configuring a cloud solution (~17.5 % of the exam)**. You must understand load balancing options, resource location availability, and Network Service Tiers.

1. Google Cloud Networking Overview

Google Cloud networking is built on Google's global private network — one of the largest in the world. Key concepts:

- **VPC (Virtual Private Cloud)** — Software-defined network that spans all GCP regions
- **Subnets** — Regional IP address ranges within a VPC
- **Firewall Rules** — Control traffic to and from instances
- **Routes** — Define paths for traffic leaving instances
- **Load Balancers** — Distribute traffic across instances

VPC Key Properties

- VPCs are **global** (span all regions)
- Subnets are **regional** (span all zones within a region)
- VPCs can be in **auto mode** (automatic subnets) or **custom mode** (manual subnets)
- Every project gets a **default VPC** with auto-mode subnets in all regions

```
VPC (Global)
├─ Subnet us-central1 (10.128.0.0/20)
│   ├── Zone us-central1-a
│   ├── Zone us-central1-b
│   └─ Zone us-central1-c
├─ Subnet europe-west1 (10.132.0.0/20)
│   ├── Zone europe-west1-b
│   ├── Zone europe-west1-c
│   └─ Zone europe-west1-d
└─ Subnet asia-east1 (10.140.0.0/20)
    ├── Zone asia-east1-a
    ├── Zone asia-east1-b
    └─ Zone asia-east1-c
```

2. Load Balancing

Google Cloud offers multiple load balancer types. Choosing the right one is a common exam topic.

Load Balancer Decision Matrix

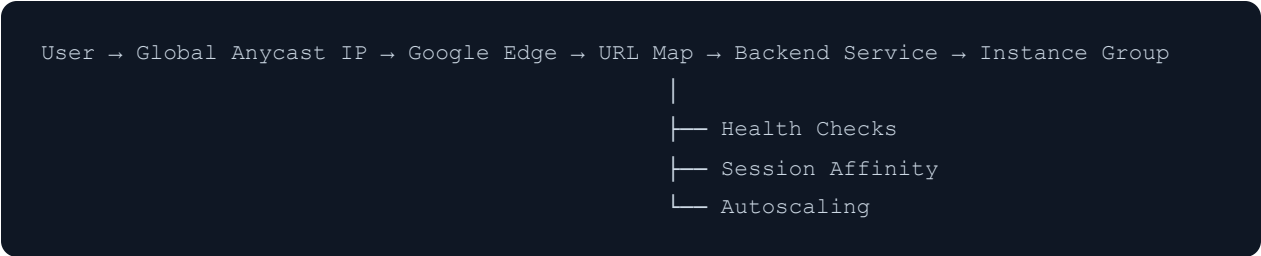
Load Balancer	Scope	Layer	Traffic Type	Use Case
External Application LB	Global	L7 (HTTP/S)	External HTTP/HTTPS	Web apps, APIs, content- based routing
Internal Application LB	Regional	L7 (HTTP/S)	Internal HTTP/HTTPS	Internal microservices, service mesh
External Proxy Network LB	Global	L4 (TCP/SSL)	External TCP/SSL	TCP/SSL termination, non-HTTP traffic
Internal Proxy Network LB	Regional	L4 (TCP)	Internal TCP	Internal TCP services
External Passthrough Network LB	Regional	L4 (TCP/UDP)	External TCP/UDP	UDP traffic, non-proxied TCP, gaming, IoT
Internal Passthrough Network LB	Regional	L4 (TCP/UDP)	Internal TCP/UDP	Internal TCP/UDP services, HA databases

External Application Load Balancer (HTTP/S LB) — Most Common on Exam

The most feature-rich load balancer, ideal for web applications.

Key Features: - **Global** — Single anycast IP, serves traffic from the nearest healthy backend - **Layer 7** — Can route based on URL path, host header, HTTP headers - **SSL termination** — Manages SSL certificates - **Cloud CDN integration** — Cache content at edge locations - **Cloud Armor integration** — DDoS protection and WAF - **URL maps** — Route `/api/*` to one backend, `/images/*` to another - **Backend services** — Instance groups, NEGs, Cloud Storage buckets

Architecture:



Configuration Components: 1. **Frontend** — IP address, port, SSL certificate 2. **URL Map** — Routing rules (host/path-based) 3. **Backend Service** — Backend groups, health checks, session affinity 4. **Backend** — Instance groups, NEGs, Cloud Storage buckets 5. **Health Check** — Determines backend health

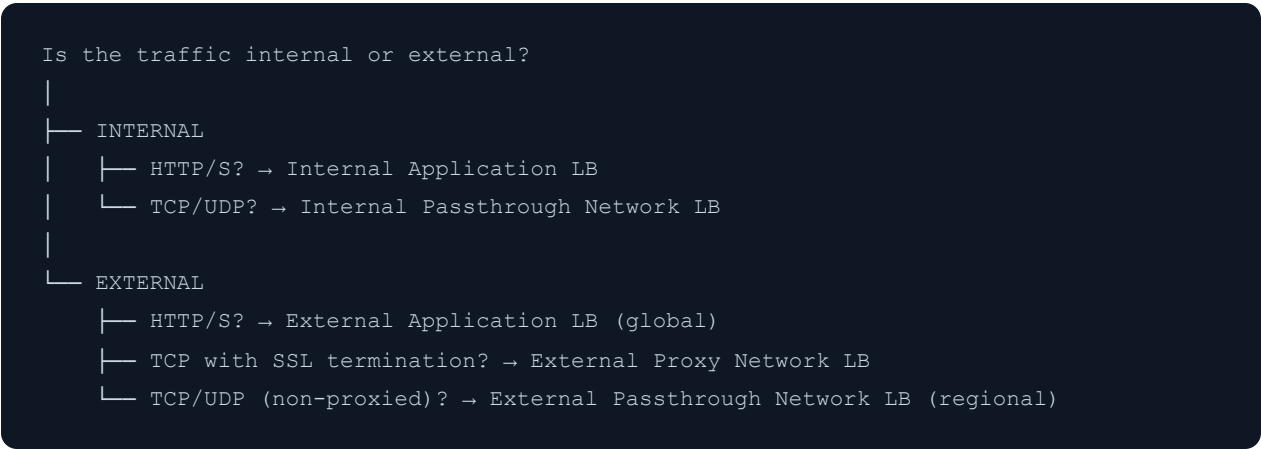
Internal Passthrough Network LB

Key Features: - **Regional** — Load balances within a single region - **Layer 4** — TCP/UDP traffic - **Internal** — Only accessible from within the VPC (or via VPN/Interconnect) - Use cases: Database HA, internal services

External Passthrough Network LB

Key Features: - **Regional** — Serves traffic to backends in a single region - **Layer 4** — TCP/UDP traffic - **Non-proxied** — Traffic passes directly to backends (preserves source IP) - Use cases: UDP gaming servers, IoT ingestion, non-HTTP protocols

Load Balancer Selection Flowchart



Health Checks

All load balancers use health checks to determine backend availability:

Protocol	Port	Interval	Example
HTTP	80	10s	Check <code>/health</code> returns 200
HTTPS	443	10s	Check <code>/healthz</code> returns 200
TCP	Any	10s	TCP connection succeeds
SSL	443	10s	SSL handshake succeeds
gRPC	Any	10s	gRPC health check

3. Cloud CDN (Content Delivery Network)

What Is Cloud CDN?

- Caches content at Google’s **edge locations** worldwide
- Reduces latency for users by serving cached content from nearby locations
- Integrated with the **External Application Load Balancer**

Key Features

- Cache static content (images, CSS, JS, videos)
- Cache keys based on URL, headers, cookies, query strings
- Signed URLs and signed cookies for access control
- Cache invalidation via API or Console
- Supports Cloud Storage buckets and instance groups as origins

When to Use Cloud CDN

- Serving static content globally
- Reducing backend load for cacheable content
- Improving page load times for global audiences

4. Resource Location Availability

Regions and Zones

Concept	Definition	Example
Region	Independent geographic area	<code>us-central1</code> (Iowa), <code>eu-west-1</code> (Belgium)
Zone	Deployment area within a region	<code>us-central1-a</code> , <code>us-central1-b</code>
Multi-region	Large geographic area (Americas, Europe, Asia)	<code>US</code> , <code>EU</code> , <code>ASIA</code>

Zone and Region Counts (approximate)

- **40+ regions** worldwide
- **3-4 zones** per region (typically)

- **Regions span:** Americas, Europe, Middle East, Africa, Asia Pacific

Resource Scope by Level

Scope	Resources
Global	VPCs, firewall rules, routes, external IP addresses, images, snapshots, load balancers (some), Cloud CDN
Regional	Subnets, regional persistent disks, regional instance groups, Cloud Router, Cloud NAT, IP addresses (internal)
Zonal	VM instances, zonal persistent disks, GKE node pools, local SSDs

High Availability Patterns

Single zone — No HA (single point of failure)

```
Zone A: [VM1] [VM2]
```

Multi-zone (within region) — Protects against zone failure

```
Zone A: [VM1] [VM3]
Zone B: [VM2] [VM4]
```

Multi-region — Protects against region failure

```
Region US-Central: [VM1] [VM2]
Region EU-West:    [VM3] [VM4]
Global LB → Routes to nearest healthy region
```

Choosing a Region — Key Factors

1. **Latency** — Choose regions close to your users
2. **Compliance** — Data residency requirements (e.g., EU data must stay in EU)
3. **Service availability** — Not all services are available in all regions
4. **Pricing** — Costs vary by region (e.g., us-central1 is typically cheapest)
5. **Carbon footprint** — Some regions run on cleaner energy
6. **Disaster recovery** — Choose geographically distant regions for DR

5. Network Service Tiers

Google Cloud offers two network service tiers that affect how traffic is routed between users and your GCP resources.

Premium Tier (Default)

```
User → Nearest Google Edge PoP → Google's Global Network → GCP Region
      (enters Google network      (private backbone,      (your resources)
      close to user)              low latency)
```

Characteristics: - Traffic enters Google's network at the **nearest edge Point of Presence (PoP)**
- Routed over Google's **private global fiber backbone** - **Lower latency**, higher reliability -
Required for **global load balancing** - **Higher cost** (but often worth it for latency-sensitive apps) -
Default tier for all resources

Standard Tier

```
User → Public Internet → Google Network Entry (near GCP region) → GCP Region
      (traverses public      (enters Google network      (your resources)
      internet longer)      near destination)
```

Characteristics: - Traffic routed over the **public internet** for most of the journey - Enters Google's network at the **region** where your resource is located - **Higher latency**, lower reliability (depends on ISP quality) - Only supports **regional load balancing** (no global LB) - **Lower cost** (good for cost-sensitive, non-latency-critical workloads)

Comparison

Feature	Premium Tier	Standard Tier
Routing	Google's global backbone	Public internet
Latency	Lower	Higher
Reliability	Higher	Variable
Global LB	Yes	No (regional only)
SLA	Google's network SLA	ISP-dependent
Cost	Higher	Lower
Default	Yes	No
Use case	Production, latency-sensitive	Dev/test, cost-sensitive

Choosing a Tier

```

Is your workload latency-sensitive or production-critical?
|
├─ YES → Premium Tier
|
└─ NO
    └─ Need global load balancing? → Premium Tier
        └─ Cost is the primary concern? → Standard Tier
  
```

6. Firewall Rules

Implied Rules

Every VPC has two implied rules that cannot be deleted: - **Default-deny all ingress** — Blocks all inbound traffic unless a rule explicitly allows it - **Default-allow all egress** — Permits all outbound traffic unless a rule explicitly denies it

Firewall Rule Components

Component	Description	Values / Example
Direction	Which way traffic flows	<code>INGRESS</code> (inbound) or <code>EGRESS</code> (outbound)
Priority	Evaluation order (lower = higher priority)	0–65535; default is 1000
Target	Which instances the rule applies to	All instances, network tags, or service accounts
Source (ingress)	Where traffic originates	CIDR ranges, tags, service accounts
Destination (egress)	Where traffic is going	CIDR ranges
Protocol/Ports	Traffic type to match	<code>tcp:80</code> , <code>udp:53</code> , <code>icmp</code> , <code>all</code>
Action	What to do with matched traffic	<code>ALLOW</code> or <code>DENY</code>

gcloud Commands

```
gcloud compute firewall-rules create RULE_NAME --network=VPC_NAME --direction=INGRESS --priority=1000
gcloud compute firewall-rules create allow-ssh-from-iap --network=default --direction=INGRESS --priority=1000
gcloud compute firewall-rules list --filter="network=my-vpc" --sort-by=priority
gcloud compute firewall-rules describe RULE_NAME
gcloud compute firewall-rules delete RULE_NAME
```

Priority Conflict Resolution

- Lower priority number wins (e.g., priority 500 overrides priority 1000)
- If two rules have the **same priority**, `DENY` wins over `ALLOW`

Exam tip: Ingress rules specify a **source**; egress rules specify a **destination**. Mixing these up is a common exam distractor.

7. VPC Peering and Shared VPC

VPC Peering

- **No transitive peering** — If $A \leftrightarrow B$ and $B \leftrightarrow C$, VPC A cannot reach VPC C through B. A direct $A \leftrightarrow C$ peering is required.

- **Both sides must create a peering connection** — Peering is only active when both networks configure it.
- No overlapping IP ranges allowed between peered VPCs.

```
gcloud compute networks peerings create PEERING_NAME --network=VPC_A --peer-project=OTHER_PROJECT_ID
gcloud compute networks peerings list --network=VPC_A
gcloud compute networks peerings delete PEERING_NAME --network=VPC_A
```

Shared VPC

- **Host project** owns the VPC and its subnets; **service projects** use subnets from the host project.
- Requires the **Shared VPC Admin** role (`roles/compute.xpnAdmin`) granted at the **organization** or **folder** level.
- Network administration is centralized; billing remains separate per project.

```
gcloud compute shared-vpc enable HOST_PROJECT_ID
gcloud compute shared-vpc associated-projects add SERVICE_PROJECT_ID --host-project=HOST_PROJECT_ID
gcloud compute shared-vpc associated-projects list --host-project=HOST_PROJECT_ID
```

Exam tip: Service project resources (VMs, GKE nodes, etc.) use **host project subnets**. IAM must explicitly grant `roles/compute.networkUser` on the shared subnet to service project service accounts.

8. Private Google Access

- Allows VM instances with **no external IP** to reach Google APIs (e.g., Cloud Storage, BigQuery) via internal IP routing — traffic stays on Google's network.
- Enabled at the **subnet level**:

```
gcloud compute networks subnets update SUBNET_NAME --region=REGION --enable-private-ip-gateway
```

- Does **NOT** allow general internet access — only routes traffic to `*.googleapis.com`.
- **For on-premises access to Google APIs:** use Private Google Access for on-premises, which requires Cloud VPN or Interconnect plus DNS configuration pointing `googleapis.com` to the restricted VIP range `199.36.153.8/30`.

6. Cloud Interconnect and VPN (Preview for Section 3)

Connectivity Options to On-Premises

Option	Bandwidth	Latency	SLA	Use Case
Dedicated Interconnect	10-200 Gbps	Lowest	99.9-99.99%	Large, predictable data transfer
Partner Interconnect	50 Mbps - 50 Gbps	Low	99.9-99.99%	Medium bandwidth needs
HA VPN	1.4-3 Gbps per tunnel	Variable	99.99%	Encrypted connection over internet
Classic VPN	1.4 Gbps per tunnel	Variable	99.9%	Legacy, simple VPN

7. Cloud DNS

What Is Cloud DNS?

- **Managed DNS service** — Authoritative DNS hosting
- 100% SLA for authoritative name serving
- Supports public and private DNS zones

Key Features

- **Public zones** — Resolve domain names from the internet
- **Private zones** — Resolve names within VPC networks (not visible to internet)
- **DNS forwarding** — Forward queries to on-premises DNS servers
- **DNS peering** — Share DNS zones between VPCs
- DNSSEC support for security

8. Cloud NAT (Network Address Translation)

What Is Cloud NAT?

- Allows VMs **without external IP addresses** to access the internet

- Provides **outbound-only** connectivity (VMs can initiate connections out, but external traffic cannot initiate connections in)
- Software-defined, **no proxy VMs** needed

Key Features

- Automatic and manual NAT IP allocation
- Configurable min/max ports per VM
- Logging support
- Regional resource

When to Use Cloud NAT

- VMs need to download packages or updates from the internet
- VMs need to call external APIs
- Security requirement: VMs should not have external IPs

Exam Practice Questions

1. **A company needs to serve a global web application with low latency. Traffic should be distributed to the nearest healthy backend. Which load balancer should they use?**
 - Answer: **External Application Load Balancer (HTTP/S)** — Global, L7, single anycast IP, routes to nearest healthy backend.
2. **You have internal microservices that communicate over HTTP within a VPC. Which load balancer is appropriate?**
 - Answer: **Internal Application Load Balancer** — Regional, L7, for internal HTTP/HTTPS traffic.
3. **Your gaming application uses UDP protocol and needs an external-facing load balancer. Which should you choose?**
 - Answer: **External Passthrough Network Load Balancer** — Regional, L4, supports UDP traffic.
4. **What is the main difference between Premium and Standard network tiers?**
 - Answer: Premium Tier routes traffic through Google's private global backbone (lower latency), while Standard Tier routes through the public internet (higher latency, lower cost). Premium supports global LB; Standard only supports regional LB.
5. **You want VMs without external IPs to access the internet for software updates. What service should you use?**

- Answer: **Cloud NAT** — Provides outbound internet access for VMs without external IPs.

6. A European company must ensure all user data stays within the EU. How should they plan their resource locations?

- Answer: Deploy resources exclusively in EU regions (e.g., `europa-west1`, `europa-west4`). Use organization policies (`gcp.resourceLocations`) to restrict resource creation to EU locations only.

7. Your application needs to handle 50 Gbps of data transfer between on-premises and GCP with the lowest possible latency. Which connectivity option is best?

- Answer: **Dedicated Interconnect** — Provides 10-200 Gbps with the lowest latency and a direct physical connection to Google's network.

Glossary

Anycast IP — A single IP address announced from multiple geographic locations; traffic is automatically routed to the nearest available endpoint. Used by the External Application Load Balancer for global routing.

Auto Mode VPC — A VPC network in which Google automatically creates one subnet per region with predefined IP ranges. Contrast with custom mode, where subnets are created manually.

Backend Service — A load balancer component that groups one or more backends (instance groups, NEGs, or buckets), defines health checks, session affinity, and balancing mode.

BYOL (Bring Your Own License) — A licensing model allowing customers to use their existing software licenses on cloud infrastructure, often requiring sole-tenant nodes for compliance.

CDN (Content Delivery Network) — A distributed network of edge servers that caches and serves content to users from locations geographically close to them, reducing latency.

CIDR (Classless Inter-Domain Routing) — A method for allocating IP addresses and routing that uses a notation like `10.0.0.0/24` to specify a network and its prefix length.

Classic VPN — A legacy GCP VPN option providing up to 1.4 Gbps per tunnel with a 99.9% SLA. Superseded by HA VPN for most use cases.

Cloud Armor — A GCP service that provides DDoS protection and Web Application Firewall (WAF) capabilities, integrated with the External Application Load Balancer.

Cloud CDN — Google Cloud's Content Delivery Network service; caches content at Google edge locations worldwide and is integrated with the External Application Load Balancer.

Cloud DNS — Google Cloud's fully managed, authoritative DNS hosting service supporting public zones, private zones, DNS forwarding, DNS peering, and DNSSEC.

Cloud Interconnect — A family of GCP services (Dedicated Interconnect and Partner Interconnect) providing private, high-bandwidth, low-latency connections between on-premises networks and Google Cloud.

Cloud NAT (Network Address Translation) — A GCP managed service that allows VM instances without external IP addresses to initiate outbound connections to the internet while preventing inbound connections from the internet.

Cloud Router — A GCP managed service that dynamically exchanges routes between a VPC and on-premises networks using BGP (Border Gateway Protocol), used with Dedicated Interconnect, Partner Interconnect, and HA VPN.

Custom Mode VPC — A VPC network where subnets are created manually, giving full control over IP ranges. Recommended for production environments.

DDoS (Distributed Denial of Service) — An attack that floods a service with traffic from multiple sources to make it unavailable. Cloud Armor provides DDoS protection.

Dedicated Interconnect — A GCP connectivity option providing a direct physical connection between on-premises infrastructure and Google's network, offering 10–200 Gbps bandwidth and the lowest latency.

Default VPC — An auto-mode VPC network automatically created in every new GCP project, with one subnet pre-configured in each region.

DNS (Domain Name System) — A hierarchical naming system that translates human-readable domain names (e.g., `example.com`) into IP addresses.

DNSSEC (DNS Security Extensions) — A set of extensions to DNS that add cryptographic signatures to DNS records, protecting against DNS spoofing and cache poisoning. Supported by Cloud DNS.

DR (Disaster Recovery) — Strategies and procedures to restore systems and data after a catastrophic event. Multi-region deployments are a common GCP DR pattern.

Edge Location (PoP — Point of Presence) — A geographically distributed network node where Google's infrastructure connects to the public internet and where Premium Tier traffic enters Google's private backbone.

Egress — Network traffic flowing outbound from a resource (e.g., from a VM to the internet). Cloud NAT handles egress for VMs without external IPs.

External Application Load Balancer — A global, Layer 7 (HTTP/S) GCP load balancer that uses a single anycast IP, supports URL-based routing, SSL termination, Cloud CDN, and Cloud Armor. Formerly called "HTTP(S) Load Balancer."

External IP Address — A publicly routable IP address assigned to a GCP resource, allowing it to communicate with the internet.

External Passthrough Network LB — A regional, Layer 4 (TCP/UDP) load balancer that forwards traffic directly to backends without proxying, preserving the client source IP. Suitable for UDP and non-HTTP protocols.

External Proxy Network LB — A global, Layer 4 (TCP/SSL) load balancer that terminates TCP/SSL connections and proxies traffic to backends.

Firewall Rule — A GCP VPC resource that permits or denies traffic to or from VM instances based on direction, protocol, ports, and source/destination identifiers.

GCP (Google Cloud Platform) — Google's suite of cloud computing services, including compute, storage, networking, databases, machine learning, and more.

gcloud — The primary command-line tool for interacting with GCP services, part of the Google Cloud SDK.

gRPC — A high-performance, open-source Remote Procedure Call (RPC) framework developed by Google. Supported as a health check protocol by GCP load balancers.

HA (High Availability) — Design approaches that minimize downtime by eliminating single points of failure, typically through redundancy across multiple zones or regions.

HA VPN — A GCP VPN offering with two redundant tunnels providing a 99.99% SLA. Supports up to ~3 Gbps aggregate throughput.

Health Check — A probe configured on a load balancer to determine whether backend instances are healthy and capable of serving traffic.

Host Project — In Shared VPC, the project that owns the VPC network and subnets; service projects attach to the host project to use its network resources.

HTTP (Hypertext Transfer Protocol) — The application-layer protocol used for web communication. Layer 7 load balancers operate at this level.

HTTPS (HTTP Secure) — HTTP encrypted with TLS/SSL, used for secure web communication. Managed SSL certificates can be provisioned by GCP.

IAM (Identity and Access Management) — GCP's system for controlling who (identity) can do what (role/permission) on which resources. Used for firewall rule targets, Shared VPC access, and more.

IAP (Identity-Aware Proxy) — A GCP service that enforces access control at the application layer, allowing secure access to GCP resources without a VPN. Used for SSH tunneling to VMs without external IPs.

ICMP (Internet Control Message Protocol) — A network-layer protocol used for diagnostic purposes (e.g., `ping`). Can be specified in GCP firewall rules.

Ingress — Network traffic flowing inbound to a resource (e.g., from the internet to a VM). Firewall rules controlling inbound traffic use `INGRESS` direction.

Internal Application LB — A regional, Layer 7 (HTTP/S) GCP load balancer accessible only from within a VPC or connected networks. Used for internal microservices.

Internal Passthrough Network LB — A regional, Layer 4 (TCP/UDP) load balancer accessible only within a VPC. Suitable for HA database configurations and internal TCP/UDP services.

Internal Proxy Network LB — A regional, Layer 4 (TCP) proxied load balancer for internal TCP traffic within a VPC.

IoT (Internet of Things) — A network of physical devices that collect and exchange data. GCP's External Passthrough Network LB is often used for IoT ingestion requiring UDP support.

IP Address — A numerical label assigned to a device on a network. GCP resources use both internal (private) and external (public) IP addresses.

ISP (Internet Service Provider) — A company that provides internet access. Standard Tier routing relies on ISP networks, which can introduce variable latency.

L4 (Layer 4) — The transport layer of the OSI model, handling TCP and UDP traffic. L4 load balancers route based on IP address and port without inspecting application content.

L7 (Layer 7) — The application layer of the OSI model, handling HTTP/S and other application protocols. L7 load balancers can route based on URL paths, hostnames, and HTTP headers.

NEG (Network Endpoint Group) — A GCP configuration object specifying a group of backend endpoints (IP:port pairs, serverless services, or internet endpoints) that can be used as backends for load balancers.

Network Service Tiers — GCP's two-tier system (Premium and Standard) controlling how traffic between users and GCP resources is routed over Google's private backbone vs. the public internet.

Organization Policy — A GCP governance feature that enforces constraints on resource configurations across an entire organization, folder, or project (e.g., `gcp.resourceLocations` to restrict deployments to specific regions).

Partner Interconnect — A GCP connectivity option that provides a connection between on-premises networks and GCP through a supported service provider, supporting 50 Mbps to 50 Gbps.

Partner Interconnect — A GCP connectivity option that provides a connection between on-premises networks and GCP through a supported service provider, supporting 50 Mbps to 50 Gbps.

PoP (Point of Presence) — See Edge Location. A network access point where Premium Tier traffic enters Google's private global backbone close to the user.

Premium Tier — The default GCP Network Service Tier that routes traffic over Google's private global fiber backbone, providing lower latency and higher reliability than Standard Tier.

Private Google Access — A subnet-level GCP setting that allows VM instances without external IP addresses to reach Google APIs and services (e.g., Cloud Storage, BigQuery) using internal IP routing.

Private Zone (DNS) — A Cloud DNS zone that resolves domain names only within specified VPC networks, not visible to the public internet.

Protocol — A set of rules governing data communication (e.g., TCP, UDP, ICMP, HTTP). Firewall rules and load balancers are configured to handle specific protocols.

Public Zone (DNS) — A Cloud DNS zone that resolves domain names for queries originating from anywhere on the internet.

Region — A specific geographic location where GCP resources are hosted (e.g., `us-central1` in Iowa). Contains multiple isolated zones.

Route — A GCP networking resource that defines the path that outbound traffic from a VM takes to reach a destination IP range.

Service Account — A special GCP account used by applications and services (rather than humans) to authenticate and authorize API calls. Can be used as a firewall rule target for fine-grained access control.

Service Project — In Shared VPC, a project that consumes subnets from a host project. Billing and resource management remain separate from the host project.

Shared VPC — A GCP networking feature allowing multiple projects (service projects) to share subnets from a centrally managed VPC (in the host project), enabling network administration centralization.

SLA (Service Level Agreement) — A commitment by GCP to a minimum level of service availability (e.g., 99.99%). Premium Tier carries Google's network SLA; Standard Tier depends on ISP quality.

SSL (Secure Sockets Layer) — A cryptographic protocol for securing network communications, now largely superseded by TLS. GCP load balancers can terminate SSL/TLS connections.

SSL Termination — The process of decrypting SSL/TLS-encrypted traffic at a load balancer before forwarding it (often unencrypted) to backend servers.

Standard Tier — A GCP Network Service Tier that routes traffic over the public internet until it reaches the GCP region where the resource resides. Lower cost but higher and more variable latency than Premium Tier.

Subnet — A regional IP address range within a VPC network. VM instances in a subnet share the same IP address block and can communicate without routing.

TCP (Transmission Control Protocol) — A connection-oriented transport protocol providing reliable, ordered delivery. Used by HTTP, HTTPS, SSH, and many other application protocols.

TLS (Transport Layer Security) — The successor to SSL; a cryptographic protocol for securing communications over networks. GCP manages TLS certificates for HTTPS load balancers.

UDP (User Datagram Protocol) — A connectionless transport protocol providing fast but unreliable delivery. Used for DNS, gaming, IoT, and streaming applications. Supported by L4 (passthrough) load balancers.

URL Map — A GCP load balancer configuration resource that defines routing rules, mapping incoming request URLs (host and path) to specific backend services.

VPC (Virtual Private Cloud) — A global, software-defined private network in GCP that provides networking for VM instances and other resources, including subnets, firewall rules, and routes.

VPC Peering — A GCP networking feature connecting two VPC networks so resources in each can communicate using internal IP addresses. Peering is non-transitive.

VPN (Virtual Private Network) — An encrypted tunnel over the public internet connecting an on-premises network to a GCP VPC. GCP offers Classic VPN and HA VPN.

WAF (Web Application Firewall) — A security system that filters and monitors HTTP traffic between a web application and the internet, protecting against common exploits. Provided by Cloud Armor.

Zone — An isolated deployment area within a GCP region (e.g., `us-central1-a`). Zonal failures affect only that zone, not the entire region.